Unit - II Capital Budgeting

Pay-back period = $\frac{\text{Cashoutlay of the project OR Original cost of The Asset}}{\text{Annual cash Inflows.}}$

1Q) A project was ₹ 1 lakh and yields and annual cash Inflows ₹ 20,000 per 8 years Calculate the pay-back period.

Yrs	Cashflow ₹
1	20,000
2	20,000
3	20,000
4	20,000
5	20,000
6	20,000
7	20,000
8	20,000

Pay-back period = $\frac{1,00,000}{20,000}$ = 5 yrs.

2Q Determine payback period for a project which requires a cash outlet of ₹ 10,000 and generate cashflows ~~of~~ 2000, ~~3000~~, 4000, 3000, 2000 1st 2nd 3rd and 4th Year respectfully.

Yrs.	Cash Inflows
1	2,000
2	4,000
3	3,000
4	2,000

Remaining = $\frac{₹1000}{₹2000}$ = 0.5 yrs = 6 mths

Total cash outlet ₹ 10,000

Total cash Inflows in the first three years = 9,000

upto the third year the total ~~cash inflows~~ cost is not recovered

but the total cash inflows for 4th year are 9000+2000 = 11,000 ₹1,000 more than the cost of project so the payback period somewhere between 3 & 4th year. Assuming that the cash inflows occurs evenly throughout the year the time required to recover ₹ 1000 will be $\frac{1000}{2000}$ = 0.5 yrs = 6 Months

Hence, the payback period is 3 years 6 months.

3Q project was ₹ 5 lakh and ~~ear~~ yields annually a profit of ₹ 80,000 after depriciation ~~tax~~ ~~g~~ at the @ 12% per annum but before tax of 50%. Calculate the payback period

Profit before tax =	₹ 80,000
Less: Tax @ 50% (80,000 × 50%)	₹ 40,000
Profit after tax	40,000
Depreciation (12% on 5,00,000) →	60,000
Annual cash Inflows → Profit before depreciation but after tax	1,00,000

pay back period

$$= \frac{500,000}{1,00,000} = 5 yrs$$

Q) There are two projects X & Y each project requires and investment of ₹ 20,000. You are required to rank these projects according to pay Back method.

Net profit Before depreciation and tax

Yrs	Project X	Project Y
1	1,000	2000
2	2,000	4000
3	4,000	6000
4	5,000	8000
5	8,000	—
	20,000	20,000

sol

The payback period project X is 5 years

(₹ 1000 + 2000 + 4000 + 5000 + 8000 = 20,000)

while the payback period for project Y is 4 years

(₹ 2000 + ₹ 4000 + ₹ 6000 + ₹ 8000 = 20,000)

Hence, project Y should be preferred or ranked first.

Q) X limited is producing articles mostly by manual labour & is considering to replace it by a new machine. There are two alternative models M & N of new machine. Calculate the payback period.

	Machine M	Machine N
estimated life of Machine	4 yrs	5 yrs
Cost of Machine	₹ 90,000	₹ 1,80,000
estimated savings in scrap	₹ 5,000	₹ 8,000
estimated savings in Direct wages	₹ 60,000	₹ 80,000
Additional cost of Maintence	₹ 8000	₹ 10,000
Additional cost of supervision	₹ 12,000	₹ 16,000

Savings - Cost = Inflows

Cost of machine = Overflow.

	M	N
a) estimated Savings p.A		
scrap	5000	8000
Direct wages	60000	80000
Total Savings		
Additional Cost p.A		
Maintenence	8000	10,000
Supervision	12000	16000
Total additional Cost (b)		
Net annual cash Inflow (a-b)	45000	60000
Cost	90000	180000
	90000	180000
	45000	60000
→	2 yrs	3 yrs

Q) Calculate net present value cost of the project @ a 21,000
b 21,000

Yr	Cash Inflow A	Cash Inflow B
1	5,000	6,000
2	5,500	7,000
3	6,000	11,000
4	6,500	4,500
5	5,500	
	28,500	28,500

P.V factor × ~~present~~ cash Inflow = P.V

Sol:-

Yr	P.V. factors at 11%	A Cash Inflow	A Present value	B cash Inflow	B present value
1	0.9009 $\frac{1}{1+0.11}$	5000	4505.5	6000	5405
2.	0.8116 $\frac{0.9009}{1.11}$	5,500	4463.8	7000	5681
3.	0.7311	6,000	4386.6	11000	8042
4.	0.6587	6,500	4281.5	4500	2964
5.	0.5934	5,500	3,263.7	–	–
Total cash Inflow		20,900			22,092
(-) cash outflow		21,000			21,000
Net present value →		-100			1,092

The net present value of project B is positive and higher than project A, project B should be selected.

Q) From the following information calculate net present value of the 2 projects & suggest which of the 2 project will be accepted. Assuming a discount rate of 10%.

	Project X	Project Y
Initial investment	₹ 20,000	₹ 30,000
estimated life	5 yrs	5 yrs
scarp value	₹1000	₹2000

	Yrs 1	2	3	4	5
project X	5000	10,000	10,000	3,000	2,000
project Y	20,000	10,000	5,000	3000	2,000

Yrs	P.V factor at 10%	X cash Inflows	X present value	Y Cash Inflow	Y present value
1	0.9090	5000	4545	20,000	18180
2	0.82644	10,000	8264	10,000	8264
3.	0.7513	10,000	7513	5,000	3758
4	0.6830	3,000	2049	3,000	2049
5	0.62099	2,000	1242	2,000	1242
+ scrap	0.620	1000	~~23601~~	2000	~~33,484~~
		Inflows -	23613		33,491
	Inflow after + scrap - Investment		24613 (-) 20,000		35491 (-) 30,000
		NPV	4613		5491

Q) A project costs of 6,00,000 & yields annually 90,000 profit after depreciation at ~~125~~ 12.5% p.A. But before tax at 50%. Calculate payback period.

Sol- Profits → 90,000
(-) tax @50% = 45,000
45,000
+ depreciation 39375
profit → 90,000

Profits = 90,000
less tax @50% = 45,000
45,000
Add dep @12.5% = 75,000 (cost × %)
1,20,000

Q) Calculate the payback period of the following projects each requiring a cash outlay of ₹ 1,00,000 suggest which project are acceptable.

Year	Cash Inflow project A	B	C
1	30,000	30,000	10,000
2	30,000	40,000	20,000
3	30,000	20,000	30,000
4	30,000	10,000	40,000
5	30,000	5000	–
	150,000	105,000	1,00,000

Project A = $\frac{1,00,000}{30,000}$ = 3.333

project B = 4 yrs

project C = 4 yrs.

Average rate of return.

$\frac{\text{Total profit (After dep \& tax)}}{\text{Net Investment in the project} \times \text{Number of years of profit}}$

(OR)

$\frac{\text{Average annual profit}}{\text{Net Investment in the project}} \times 100$

Q) A project requires an Investment of ₹ 5,00,000 & has a scrap value of ₹ 20,000 after 5 yrs. It is expected to yield profit after dep & tax. During the 5 years amount in the rupees 40,000, 60,000, 70,000, 50,000 & 20,000. Calculate the average rate of return on the investment.

P.T.O

$$\frac{\text{Avg Annual profits}}{\text{net Investment}} \times 100$$

$$\text{Avg Annual profits} = \frac{2,40,000}{5}$$

$$= 248,000$$

$$= \frac{248,000}{5,00,000 - 20,000} \times 100$$

$$= \frac{48,000}{4,80,000} \times 100$$

= 10% average rate of return

7) No project is acceptable unless the yield is 10% th cash Inflows of a certain project along cash outflows are given below

Years	Outflow	Inflow
0	1,50,000	—
1	30,000	20,000
2		30,000
3		60,000
4		80,000
5		30,000
5th		40,000

The Salvage value at the end of 5th year is 40,000. Calculate the present value.

Calculation of Present Value of cash Inflow.

Yr	Inflow	PVF @10%	Presentvalue
1	20,000	0.909	18,180
2	30,000	0.826	24,780
3	60,000	0.751	45,060
4	80,000	0.683	54,640
5	30,000	0.620	18,600
5	40,000	0.620	24,800
	P.v of annual cash Inflow → 1,86,060		

Cash calculation of present value of Cash outflow

Yr	Outflow	PVF @10%	Present value
0	1,50,000	1.	1,50,000.
1	30,000	0.909	27,270
	P.V of cash outflow → 1,77,270		

Calculation of net present value - P.V of Cash Inflow 1,86,060

(-) P.V of cash outflow 1,77,270

Net present value - 8,790

Q) A Company is considering investment in a project that Costs ₹ 2,00,000. The project has an expected life of 5 yrs and zero ~~salvage~~ Salvage value. The Company uses straight line method of depreciation. The company's tax rate is 40%.

The Estimated earnings before depreciation and before tax from the project are as follows.

Yr	Earning before depreciation and tax	Presented value factor @10%
1	70,000	0.909
2	80,000	0.826
3	1,20,000	0.751
4	90,000	0.683
5	60,000	0.621

You are required to calculate the net present value at 10% and advise the Company.

Sol Yr	b Earning before dep & Tax	c Dep 2,00,000/5 yrs	d Earning before tax (b-c)=d	e = d × tax % → 40% @40%	(f=d-e) Earning after TAX (EAT)	(g=f+c) Cashflow (EAT+Dep)
1	70,000	40,000	30,000	12,000	18,000	58,000
2	80,000	40,000	40,000	16,000	24,000	64,000
3	1,20,000	40,000	80,000	32,000	48,000	88,000
4	90,000	40,000	50,000	20,000	30,000	70,000
5	60,000	40,000	20,000	8,000	12,000	52,000

Hall Ticket No: 1214234008012

ST. JOSEPH

PVF@10%	Present values → PVF × cashflow
0.909	52,722
0.826	52,864
0.751	66,083
0.683	47,810
0.621	32,292
	2,51,776 → P.V of cash Inflows.
	2,00,000 → P.V of cash outflow
(-)	51,776 → net present value.

Outflow - we spent
Inflow - we got / Income
negative Impact - when outflow is more
positive Impact - when Inflow is more.

Q) present value factor @ ~~10~~ 15% per annum.
Interest rate = 5,00,000

Yr	Net cash flows	PVF@15%	
1.	Nil	0.87	0.87
2	100,000	0.76	0.76
3	160,000	0.66	0.66
4	240,000	0.57	0.57
5	300,000	0.50	0.50
6	600,000	0.43	0.43

P.V Total
− Interest
net Present

Yr	Net Cash flow	PVF @15%	P.V
1	–	0.87	0.87
2	1,00,000	0.76	76,000
3	1,60,000	0.66	1,05,600
4.	2,40,000	0.57	1,36,800
5	3,00,000	0.50	1,50,000
6.	6,00,000	0.43	2,58,000
	PV of cash Inflow		7,26,400
	PV of cash outflow		5,00,000
	Npv		2,26,400

Internal rate of return

Q) Initial Investment = 60,000, life of assets 4 yrs,

life of assets = 4 yrs

estimated annual cashflows = ~~15000~~.

Yrs	Annual cashflow
1	15,000
2	20,000
3	30,000
4	20,000

Yr	Annual Cashflow	PVF @10	P.V	PVF @12%	PV	PVF @14%	PV	PVF @15%	PV
1	15,000	0.909	13,635	0.892	13,380	0.877	13,155	0.869	13635
2	20,000	0.826	16,520	0.797	15,940	0.769	15,380	0.756	15,120
3	30,000	0.751	22,530	0.711	21,330	0.674	20,220	0.657	19,710
4	20,000	0.683	13,660	0.635	12,700	0.592	11,840	0.571	11,420
					63,350		60,595		59,285
	P.V of cash inflow		66,345		(-) 60,000		(-) 60,000		(-) 60,000
	P.V of cash outflow		60,000		3,350		595		-715
			6,345						

Formula

$$IRR = \text{Lower Rate} + \frac{\text{NPV of lower}}{\text{NPV of lower} - \text{NPV of Higher}} (\text{Higher rate} - \text{Lower rate})$$

$$= 14\% + \frac{595}{595-(-715)}(15-14)$$

$$= 14 + \frac{595}{595+715} \times 1 = 14 + \frac{595}{1315} \times 1 = 14 + 0.45 = 14.45\%$$

Q. Calculate Internal rate of return for the project 'x'. The details of the project are as under:

Initial Cost = 21,000

Cash Inflows (Year wise)

Yr	1	2	3	4
₹	4000	6000	8000	10,000
PVF@10%	0.909	0.826	0.751	0.683
PVF@12%	0.893	0.797	0.712	0.636

Sol:-

Yr	Cash flow	PVF @10%	PV	PVF@ 12%	PV
1	4000	0.909	3,636	0.893	3572
2	6000	0.826	4,956	0.797	4782
3	8000	0.751	6008	0.712	5696
4	10,000	0.683	6,830	0.636	6360
					20410
P.V of cash Inflow			21430		
P.V of cash outflow			(-) 21000		(-) 21000
			430		-590

$$IRR = \text{Lower rate} + \frac{\text{NPV of lower}}{\text{NPV of lower} - \text{NPV of Higher}} (\text{Higher rate} - \text{lower rate})$$

$$= 10 + \frac{430}{430-(-590)}(12-10)$$

$$= 10 + \frac{430}{430+590} \times 2$$

$$= 10 + \frac{430}{1020} \times 2$$

$$= 10 + 0.421 \times 2$$

$$= 10 + 0.842$$

$$= 10.84\%$$

1 Profitability Index = $\frac{\text{Present value of cash Inflow}}{\text{Present value of cash outflow}}$.

Q) Calculate PI from the following

Project cost = 10,00,000

Cash Inflow.

1 yr - 2,80,000

2 yr - 3,60,000

3 yr - 4,00,000

4 yr - 5,00,000

Cost of capital is 10%.

Sol.

Yr	Inflow.	Prf@10%	PV cash Inflow
1	2,80,000	0.909	254520
2	3,60,000	0.826	297360
3	4,00,000	0.751	300400
4	5,00,000	0.683	341500
			11,93,780

$$PI = \frac{11,93,780}{10,00,000}$$

$$= 1.19378$$